

Total No. of Questions : 4]

SEAT No. :

P8978

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Oct-22/BE/Insem-62

B.E. (Mechanical Engineering)

PRODUCT DESIGN AND DEVELOPMENT

(2019 Pattern) (Semester-VII) (Elective - IV) (402045 A)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Explain product development process. **[7]**

b) Explain Morphology of product design by ASIMOW. **[8]**

OR

Q2) a) Write a short note on Product Life Cycle. **[6]**

b) Explain Engineering Design process. **[5]**

c) Explain the reasons for product failure in brief. **[4]**

Q3) a) Explain Technology of S Curve with suitable example. **[7]**

b) Explain QFD (Quality Function Deployment) approach of modern product development. **[8]**

OR

Q4) a) Explain Customer need Models in brief. **[6]**

b) Explain market segmentation in detail with suitable example. **[5]**

c) Illustrate the philosophy of TRIZ methodology. **[4]**

